

1 Class Questions

Turn each into If P then Q.

1. A matrix is invertible provided that its determinant is not zero.
2. For a function to be continuous, it is sufficient that it is differentiable.
3. For a function to be integrable, it is necessary that it is continuous.
4. A function is rational if it is a polynomial.
5. An integer is divisible by 8 only if it is divisible by 4.
6. Whenever a surface has only one side, it is non-orientable.
7. A series converges whenever it converges absolutely.
8. A geometric series with ratio r converges if $|r| < 1$
9. A function is integrable provided the function is continuous.
10. The discriminant is negative only if the quadratic equation has no real solutions.
11. You fail only if you stop writing. (Ray Bradbury)
12. People will generally accept facts as truth only if the facts agree with what they already believe. (Andy Rooney)
13. Whenever people agree with me I feel I must be wrong. (Oscar Wilde)

1.1 Breakout Sessions

If $m \in \mathbf{Z}$ is even, then m^2 is even.

Proof? We know that by the definition of an even integer, there exists some $n \in \mathbf{Z}$ such that $m = 2n$. So if m^2 is an even integer, then $m^2 = 2n$, if so, $m = \sqrt{2n}$, but $\sqrt{2}$ is never an integer, so our supposition is false.

Where did we go wrong?

2 Proof Examples

Proof by cases: Prove Theorem: If q is not divisible by 3, then q^2 divided by 3 leaves 1 remaining.

q is not divisible by 3, so it either leaves 1 or 2 remaining.

case 1: $q = 3k + 1$ for some $k \in \mathbf{Z}$ so

$$\begin{aligned} q^2 &= (3k + 1)^2 = 9k^2 + 6k + 1 \\ &= 3(k^2 + 2k) + 1 \end{aligned}$$

So q^2 is divisible by 3 with a remainder of 1

case 2: $q = 3k + 2$ for some $k \in \mathbf{Z}$ so

$$\begin{aligned} q^2 &= (3k + 2)^2 = 9k^2 + 12k + 4 \\ &= 9k^2 + 12k + 3 + 1 \\ &= 3(3k^2 + 4k + 1) + 1 \end{aligned}$$

So q^2 is divisible by 3 with a remainder of 1

So in either case q^2 is divisible by 3 with a remainder of 1

2.1 Proof Examples

Prove: If q^2 is divisible by 3, so is q .

Use contrapositive, if q is not divisible by 3 then q^2 is not divisible by 3. Theorem above tells us that.